Bachelor's Thesis Topics E&OR
Spring 2026

This file contains a list of bachelor thesis topics that are proposed by members of the Department of Quantitative Economics. You are asked to carefully read these and prepare a list of your five favorite topics. It can be useful to browse trough some of the references to have a feeling about the type of literature that you will need to read. Some topics can potentially belong to more than one discipline, so have a look at all the topics and not just those from one discipline.

In case you want to work on a thesis topic developed by yourself, we recommend to contact a supervisor of the respective discipline and ask him/her whether he/she is willing to supervise your thesis and agrees with the topic.

On Canvas you will find a survey where you can submit your preferred topics. Please enter the topic numbers from this document in the corresponding fields of the first question using the format X.YY (e.g. 2.05).

If all your topic choices come from the same discipline, please also indicate an alternative discipline for your thesis topic (section number from 1,2,3,4 is enough). This is hopefully only a very rare exception in case the demand for topics from the different disciplines is very unusually distributed.

If you have developed your own topic proposal, please write your topic title and a short description of the topic in the comment field of the second question. There you can also leave other comments if applicable.

As mentioned during the info session, we try to allocate the topics in such a way that every student can work on one of their preferred topics (and ideally on one that is higher up on the list). Please keep in mind that this may not be possible at all times and that there are also some other constraints (e.g. a balance among supervisors).

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1 Actuarial Sciences

1.1 Measuring Preferences of Pension Fund Participants

Supervisor: Antoon Pelsser
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Discipline: Actuarial Sciences
Required Courses: EBC2122 Actuarial Mathematics

Description

Dutch pension funds are required to measure the preferences (or risk-aversion) of participants regarding investment risk and or the pension that can be attained at retirement.

There is a lot of attention in the economic literature on Prospect Theory, which models utility functions including loss-aversion. Make a literature study of finding good methods for soliciting preferences from pension participants. Based on this literature overview, make recommendations on how to apply the best method in a pension setting, with retirement many years (even decades) into the future.

References

Personal Life Events and the Stability of Preferences, 2023, Paul Bokern, Arno Riedl, Hans Schmeets, Jona Linde, Peter Werner, Netspar dicussion paper DP 10/2023-047, www.netspar.nl

1.2 Applying Graphical Causal Models for Financial Risk Analysis

Supervisor: Nalan Basturk, Tim Lindner
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Discipline: Actuarial Sciences
Required Courses: EBS2072 Introduction to Software in Econometrics

Description

Financial risk analysis is critical for assessing the stability and sustainability of financial systems. While existing literature frequently uses logit models to estimate risks and evaluates model accuracy through Value at Risk (VaR) violations, these methods may fail to capture the full complexity of interdependencies between financial variables. This thesis proposes the use of graphical causal models (GCMs) as an alternative tool for understanding and analyzing financial risks. GCMs offer a more nuanced approach by modeling causal relationships among variables, potentially enhancing the accuracy and interpretability of financial risk assessments. Additionally, this research aims to explore the novel aspect of assessing the accuracy of GCMs, focusing on how these models perform in the context of financial risk, particularly when compared to traditional approaches.

The main aim is to apply GCMs to examine how GCMs can model complex causal relationships between financial variables, such as asset prices, interest rates, and credit risks. The application will specifically focus on model accuracy in the context of Value at Risk (VaR) violations. The application of GCMs to real-world financial data will demonstrate these models' ability improve risk assessment and inform decision-making processes in actuarial practice. The thesis would ideally

evaluate the advantages and limitations of graphical causal models in comparison to conventional logit models, with an emphasis on assessing model accuracy using VaR and other risk metrics.

References

Probabilistic risk analysis: foundations and methods. Bedford, Tim, and Roger Cooke. Cambridge University Press, 2001.

Graphical models, causal inference, and econometric models. Spirtes, P. (2005). Journal of Economic Methodology, 12(1), 3-34.

Graphical models, causality, and intervention. Pearl, J. (2011).

Causal inference using graphical models with the R package pcalg.. Kalisch, M., Mächler, M., Colombo, D., Maathuis, M. H., & Buhlmann, P. (2012). Journal of statistical software, 47, 1-26.

1.3 Improving Credit Risk Analysis Using Survival Models with Variable Selection

Supervisor: Nalan Basturk
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Discipline: Actuarial Sciences, Econometrics
Required Courses: EBS2072 Introduction to Software in Econometrics

Description

Accurate credit risk analysis is critical for financial institutions in assessing the likelihood of default and managing loan portfolios effectively. While survival models, such as Cox proportional hazards and parametric models, have demonstrated their potential in predicting time-to-default, existing literature has not explored fully the use of variable selection techniques to enhance these models. This thesis aims to explore the application of survival models for credit risk analysis while integrating variable selection methods to identify the most important predictors of default. By improving model accuracy and interpretability through systematic variable selection, this research seeks to address a gap in the literature and provide a more robust framework for credit risk prediction.

The main objectives of this research are: to explore the use of survival models, such as Cox proportional hazards and parametric models, in predicting credit default and understanding time-to-default dynamics and to integrate variable selection techniques. The thesis will then compare the predictive performance of survival models with and without variable selection, using simulated or real datasets.

References

Gepp, A., & Kumar, K. (2008). The role of survival analysis in financial distress prediction. International research journal of finance and economics, 16(16), 13-34.

Pierri, F., & Caroni, C. (2017). Bankruptcy prediction by survival models based on current and lagged values of time-varying financial data. Communications in Statistics: Case Studies, Data Analysis and Applications, 3(3-4), 62-70.

Gao, F., & He, X. (2021). Survival analysis: theory and application in finance. In Handbook of Financial Econometrics, Mathematics, Statistics, and Machine Learning (pp. 4087-4118).

Djeundje, V. B., & Crook, J. (2019). Dynamic survival models with varying coefficients for credit risks. European Journal of Operational Research, 275(1), 319-333.

1.4 Solidarity in Pension Funds

Supervisor:Luke Servat
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Discipline: Actuarial Sciences
Required Courses: EBC2122 Actuarial Mathematics

Description

The pension system of the Netherlands is transitioning to a defined contribution plan. Due to the absence of guarantees and strong dependence on market returns in this new system, pension funds have been given the novel task of preventing the occurrence of "unlucky generations". In this thesis, we want to assess the effectiveness of several risk-sharing mechanisms to reduce differences between generations.

References

Hogere en stabielere pensioenen dankzij solidariteitsreserve. De Groot, C., & Bakker, M. , 2022, Ortec Finance.

Garanties in het nieuwe pensioenstelsel. Kokken, M., Nijman, T. E., & Werker, B. J., 2024. , Netspar.

Intergenerational Transfers in the New Dutch Pension Contract. van Bilsen, S., Mehlkopf, R.J. & van Stalborch, S., 2022. , De Economist.

2 Econometrics

2.1 Autoregressive Models for Matrix-Valued Time Series

Supervisor: Ivan Ricardo
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Discipline: Econometrics
Required Courses: EBC2120 Econometric Methods II, EBS2072 Introduction to Software in Econometrics

Description

I can potentially supervise two bachelor students interested in the analysis of matrix-valued time series. Informally, a matrix-valued time series is a dataset where, at each time point, you observe an entire matrix – for example, a table showing employment in different industries across US states each month, or house prices across states and income groups observed over time. This produces a three-dimensional data array (time $\times$ rows $\times$ columns). Traditional time series approaches usually convert this object into a matrix (e.g., time $\times$ aggregated rows and columns) or treat each series separately and therefore ignore the matrix structure; these projects will investigate whether preserving the matrix form is advantageous in an autoregressive framework.

There are several avenues which can be undertaken by the students, the general idea being that two students have different topics for investigation. Of course, these topics are not fixed, and students are welcome to have a discussion with me to see what might fit their interests best.

- Sectoral unemployment dynamics in the United States
  - Using data on the employment of different sectors of the economy (e.g., manufacturing, trade, leisure/hospitality) for different states of the USA, we may investigate how different sectors and states interact with one another dynamically.
- House prices and income in the United States (yes I know these topics are America-centric)
  - Does the ability to buy a house depend on where you live? We can try investigating the relation between house prices and income in the USA on a state level. The dataset of interest would thus be house prices and personal income for different US states over time.
- Connections of the matrix autoregressive (MAR) model to the reduced rank model
  - The MAR can be seen as a regression with a reduced rank coefficient. This project clarifies this relationship and explains what it implies for modelling matrix-valued time series.

The first two topics are more applied, while the last is more theoretical. Be aware that good programming skills are required (R, Python, or Julia).

References

Autoregressive Models for matrix-valued time series Rong Chen, Han Xiao, Dan Yang, 2021, Journal of Econometrics.

The long-run relationship between house prices and income: evidence from local housing markets Joshua Gallin, 2021, Real Estate Economics.

Sectoral shifts and cyclic unemployment David M. Lilien, 1982, Journal of Political Economy.

2.2 Bubbles in commodity prices/crypto currencies

Information

Supervisor: Alain Hecq and Lison Christiaens
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Discipline: Econometrics
Required Courses: EM1

Summary

This project aims at comparing the behavior of usual autoregressive models, e.g. an AR(1) $Y_t = \alpha_1 Y_{t-1} + \varepsilon_t$ with a new class of noncausal models with leads $Y_t = \alpha_2 Y_{t+1} + \varepsilon_t$ - as well as with the general mixed causal noncausal models, namely $(1-\alpha_1 L)(1-\alpha_2 L^{-1})Y_t = \varepsilon_t$.

Description/specific questions

Usual autoregressive models are not able to reproduce the bubble burst pattern observed in economic time series (see Hecq et al, 2016, Gourieroux and Zakoian, 2017). Mixed causal noncausal models (denoted $MAR(r,s)$) do so. The first part of the project consists in understanding the consequences of the different mixed causal and noncausal specifications, and in particular the need to depart from the Gaussian hypothesis. We have developed R routines to estimate those models, assuming a Student's t distribution.

The thesis consists in estimating mixed causal and noncausal models on several commodity/crypto currency prices after a careful investigation of the choice of the fundamental component, say for an observed price series P_t

$$\begin{aligned} P_t &= fundamental + Y_t. \\ (1-\alpha_1 L)(1-\alpha_2 L^{-1})Y_t &= \varepsilon_t \end{aligned}$$

It is not clear what fundamental component to be considered for commodities. Hencic and Gourieroux and (2015) use a cubic trend for extracting the $fundamental$ component. However alternative technical detrending strategies exist (HP filter, Beveridge Nelson trend, see Hecq and Voisin 2021), cubic splines as well as the use of "economic" fundamentals.

The first important question of this topic is to measure the fundamental part of commodities or crypto currencies (if needed when they are not stationary) to extract the stationary bubble component. The second question is to investigate the statistical features of Y_t in terms of causal and noncausal dynamics and their robustness to the type of fundamental filtering.

References

Fries, S., & Zakoian, J.-M. (2019). Mixed causal-noncausal processes and the modelling of explosive bubbles, Econometric Theory.

Hecq, A., Lieb, L. and Telg S. (2016), Identification of Mixed Causal-Noncausal Models in Finite Samples, Annals of Economics and Statistics, 123-124.

Hecq, A., Lieb, L., & Telg, S. (2017). Simulation, Estimation and Selection of Mixed Causal-Noncausal Autoregressive Models: The MARX Package. (Social Science Research Network). SSRN. More information about this publication. The MARX R package to estimate and simulate mixed causal-noncausal models is available on CRAN: https://cran.r-project.org/web/packages/MARX/index.html.

Hecq, A. and E. Voisin (2020), Forecasting bubbles with mixed causal-noncausal autoregressive models, Econometrics and statistics.

Hecq, A. and E. Voisin (2021), Predicting bubble bursts in oil prices during the COVID-19 pandemic with mixed causal-noncausal models, forthcoming in Advances in Econometrics in honor of Joon Y. Park.

Hencic, A., and C. Gourieroux (2015). Noncausal autoregressive model in application to bitcoin/usd exchange rates. In Econometrics of risk (pp. 17{40).

Gourieroux, C. and Zakoian, J.M. (2017), Local Explosion Modelling by Noncausal Process, Journal of the Royal Statistical Society, Series B, doi:10.1111/rssb.12193.

2.3 Common Bubbles in commodity prices/crypto currencies

Information

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Discipline: Econometrics
Required Courses: EM1

Summary

This project aims at comparing the behavior of usual autoregressive models, e.g. an AR(1) $Y_t = \alpha_1 Y_{t-1} + \varepsilon_t$ with a new class of noncausal models with leads $Y_t = \alpha_2 Y_{t+1} + \varepsilon_t$ - as well as with the general mixed causal noncausal models, namely $(1 - \alpha_1 L)(1 - \alpha_2 L^{-1})Y_t = \varepsilon_t$.

Description/specific questions

Usual autoregressive models are not able to reproduce the bubble burst pattern observed in economic time series (see Hecq and Voisin, 2020, Gourieroux and Zakoian, 2017). Mixed causal noncausal models (denoted $MAR(r, s)$) do so. The first part of the project consists in understanding the consequences of the different mixed causal and noncausal specifications, and in particular the need to depart from the Gaussian hypothesis. We have developed R routines to estimate those models, assuming a Student's t distribution.

The thesis consists in estimating mixed causal and noncausal models on several commodity prices after a careful investigation of the choice of the fundamental component, say for an observed price series P_t

$$\begin{aligned} P_t &= fundamental + Y_t. \\ (1-\alpha_1 L)(1-\alpha_2 L^{-1})Y_t &= \varepsilon_t \end{aligned}$$

It is not clear what fundamental component to be considered for commodities. Hencic and Gourieroux and (2015) use a cubic trend for extracting the $fundamental$ component. However alternative technical detrending strategies exist (HP filter, Beveridge Nelson trend, see Hecq and Voisin 2021), cubic splines as well as the use of "economic" fundamentals. For this topic the detrending is not so important. Let us consider the HP fitered series.

The central point is to search for combination of bubble series that don 't have the bubble feature. For instance although Y_{1t} and Y_{2t} are 2 noncausal series following a bubble pattern, does it exist a combination $X_t = Y_{1t} - \beta Y_{2t}$ that does not have the bubble feature.

References

Fries, S., & Zakoian, J.-M. (2019). Mixed causal-noncausal processes and the modelling of explosive bubbles, Econometric Theory.

Hecq, A., Lieb, L. and Telg S. (2016), Identification of Mixed Causal-Noncausal Models in Finite Samples, Annals of Economics and Statistics, 123-124.

Hecq, A., Lieb, L., & Telg, S. (2017). Simulation, Estimation and Selection of Mixed Causal-Noncausal Autoregressive Models: The MARX Package. (Social Science Research Network). SSRN. More information about this publication. The MARX R package to estimate and simulate mixed causal-noncausal models is available on CRAN: https://cran.r-project.org/web/packages/MARX/index.html.

Hecq, A. and E. Voisin (2020), Forecasting bubbles with mixed causal-noncausal autoregressive models, Econometrics and statistics.

Hecq, A. and E. Voisin (2021), Predicting bubble bursts in oil prices during the COVID-19 pandemic with mixed causal-noncausal models, forthcoming in Advances in Econometrics in honor of Joon Y. Park.

Hencic, A., and C. Gourieroux (2015). Noncausal autoregressive model in application to bitcoin/usd exchange rates. In Econometrics of risk (pp. 17{40).

Gourieroux, C. and Zakoian, J.M. (2017), Local Explosion Modelling by Noncausal Process, Journal of the Royal Statistical Society, Series B, doi:10.1111/rssb.12193.

2.4 Green bubbles and the probability of crash

Information

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Discipline: Econometrics
Required Courses: EM1

Summary

Recently the term known as the green bubble emerged in the literature. It refers to a situation where the world is overinvesting in renewable energy sources. Excessive market behavior is detrimental not only in terms of financial stability but also to the credibility of the transition process. This project aims at understanding the existence of such green bubbles (and to build such indicators).

Description/specific questions

Indeed, it is shown recently that most climate, financial indicators and by construction the green bubble (e.g. RENIXX indicator in Giancaterini et al (2026)) are non-fundamentals. This means that economic agents have more information than we econometricians. Noncausal models (see Hecq and al. since 2016 at www.alainhecq.eu) have been developed to capture such nonlinearities in time series.

Usual autoregressive models are not able to reproduce the bubble burst pattern observed in economic time series (see Hecq and Voisin, 2020, Gourieroux and Zakoian, 2017). Mixed causal noncausal models (denoted $MAR(r,s)$) do so. The first part of the project consists in understanding the consequences of the different mixed causal and noncausal specifications, and in particular the need to depart from the Gaussian hypothesis. We have developed R routines to estimate those models but using $\alpha-$stable distribution instead we can estimate the probabilities that the series collapses when a high value is reached. When $u_T \to \infty$ (namely a very high value on the bubble), then

- $u^*_{T+h} \to 0$ (crash) with probability $1-\psi^{\alpha h}$
- $u^*_{T+h} \to \psi^{-h} u_T$ (continue to increase) with probability $\psi^{\alpha h}$

This project aims at comparing the behavior of usual autoregressive models, e.g. an AR(1) $Y_t = \alpha_1 Y_{t-1} + \varepsilon_t$ with a new class of noncausal models with leads $Y_t = \alpha_2 Y_{t+1} + \varepsilon_t$ - as well as with the general mixed causal noncausal models, namely $(1-\alpha_1 L)(1-\alpha_2 L^{-1})Y_t = \varepsilon_t$. Then forecasting bubble bursts migh be investigated.

References

Gourieroux, C. and Zakoian, J.M. (2017), Local Explosion Modelling by Noncausal Process, Journal of the Royal Statistical Society, Series B, doi:10.1111/rssb.12193.

Hecq, A., Lieb, L. and Telg S. (2016), Identification of Mixed Causal-Noncausal Models in Finite Samples, Annals of Economics and Statistics, 123-124.

Hecq, A. and E. Voisin (2020), Forecasting bubbles with mixed causal-noncausal autoregressive models, Econometrics and statistics.

Giancaterini, F. and Hecq, A., Jasiak, J, and A. Nayazi (2026). Bubble Detection with Application to Green Bubbles: A Noncausal Approach.

Fries, S. (2022). Conditional moments of noncausal alpha-stable processes and the prediction of bubble crash odds. Journal of Business & Economic Statistics 40(4), 1596–1616.

2.5 Forecasting realized volatility

Information

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Discipline: Econometrics
Required Courses: EM1

Summary

This project aims at investigating realized volatility models and to compare the forecasting perforfance of different approaches.

Description/specific questions

In EM1 we mainly focuses on conditional mean models, mispecifications such as heteroskedasticity affecting the results of our estimators and inference. This topic is about modeling the second moments of a time series, namely the conditional variance that is often associated with the volatility. In the ARCH/GARCH literature this amounts to estimate a second equation to describe time varying volatility. Consider r_t the excess stock return (in deviation from its conditional mean), a GARCH(1,1) is

$$\begin{aligned} r_t &= \varepsilon_t \\ \varepsilon_t &= h_t u_t \quad u_t \sim N(0,1) \\ h_t &= \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1} \end{aligned}$$

Instead the realized volatility directly estimates the underlying volatility. Both the computational simplicity and theoretical foundations make realized volatility measures (realized variance, bi-power variation, median realized variance, etc) very attractive among practitioners and academics for modelling time varying volatilities and monitoring financial risk. As an example, the 5-minute realized variance (RV5), a benchmark often considered in practice (see Liu *et al.*, 2015), is obtained as $RV5_t \equiv \sum_{j=1}^{M} r_{j,t}^2$,where $r_{j,t} = \ln P_{j,t} - \ln P_{j-1,t}$ are the high frequency returns observed for $j = 1...M$ intra-day 5-min periods.

Corsi (2011) has proposed a simple model, called HAR, to forecast realized volatilities. The Heterogeneous Autoregressive model (VHAR) reads as follows

$$rv_{t+1}^{(d)} = \gamma + \phi^{(d)} rv_t^{(d)} + \phi^{(w)} rv_t^{(w)} + \phi^{(m)} rv_t^{(m)} + \varepsilon_{t+1}, \tag{1}$$

where (d), (w), and (m) denote, respectively, time horizons of one day, one week (5 days), and one month (22 days),

$$rv_t^{(w)} = \frac{1}{5}\sum_{i=0}^{4} rv_{t-i}^{(d)}, \quad rv_t^{(m)} = \frac{1}{22}\sum_{i=0}^{21} rv_{t-i}^{(d)},$$

Following *inter alia*, Bubák *et al.* (2011), Cubadda et al. (2017) and Souček and Todorova (2013), the joint dynamics of seveal series $a_t^{(d)} = (rv_{1t}^{(d)}, ...rv_{Kt}^{(d)})'$ can be analyzed through a multivariate version of the univariate HAR model. The Vector Heterogeneous Autoregressive model

(VHAR) reads as follows:

$$a_{t+1}^{(d)} = \gamma + \Phi^{(d)} a_t^{(d)} + \Phi^{(w)} a_t^{(w)} + \Phi^{(m)} a_t^{(m)} + \varepsilon_{t+1}, \tag{2}$$

where (d), (w), and (m) denote, respectively, time horizons of one day, one week (5 days), and one month (22 days),

$$a_t^{(w)} = \frac{1}{5} \sum_{i=0}^{4} a_{t-i}^{(d)}, \quad a_t^{(m)} = \frac{1}{22} \sum_{i=0}^{21} a_{t-i}^{(d)},$$

γ is a vector of N constant terms, $\Phi^{(i)}$ is a $N \times N$ matrix for $j = d, w, m$, and ε_t are i.i.d. innovations with $\mathrm{E}(\varepsilon_t) = 0$, $\mathrm{E}(\varepsilon_t \varepsilon_t') = \Sigma$ (positive definite), and finite fourth moments.

The aim of this research is to estimate univariate and multivariate HAR models and compare their forecasting performance.

References

[1] Bauwens, Luc ; Hafner, Christian ; Laurent, Sébastien "Volatility models", coredp2011_58web.pdf

[2] Corsi, F. (2009), A Simple Approximate Long-Memory Model of Realized Volatility, *Journal of Financial Econometrics*, 7, 174-196.

[3] Corsi, F., Audrino, F. and R. Renò (2012), HAR Modeling for Realized Volatility Forecasting, in Bauwens, L., Hafner, Ch. and S. Laurent (eds), *Handbook of Volatility Models and Their Applications* Wiley, Ch. 15.

[4] Corsi, F., S. Mittnik, C. Pigorsch, and U. Pigorsch (2008), The volatility of realized volatility, *Econometric Reviews*, 27, 46-78.

[5] Cubadda, Guardabasio and Hecq (2017), A vector heterogeneous autoregressive index model for realized volatility measures, International Journal of Forecasting.

[6] M. Souček et al. (2013), Realized volatilities transmission between crude oil and equity futures markets: A multivariate HAR approach Energy Economics.

[7] V. Bubak et al. (2011), Volatility transmission in emerging European exchange markets, Journal of Banking and Finance

2.6 The Pump-Price Effect: Energy Prices, Expectations, and Inflation

Supervisor: Enrico Wegner
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Discipline: Econometrics
Required Courses: EBC2120 Econometric Methods II, EBS2072 Introduction to Software in Econometrics

Description

Energy-price shocks are central to current macro policy debates. After the recent inflation surge and cost-of-living crisis, central banks must judge when to "look through" energy-driven inflation and when to act to prevent broader persistence that may arise if inflation expectations move. Box B in the Bank of England's November 2025 Monetary Policy Report studied, for the UK, how energy shocks transmit through inflation expectations and posed a sharp mechanism: because petrol and gas prices are highly salient, households may update expectations more strongly, potentially amplifying the macroeconomic effects of energy shocks. In this thesis, you will take this idea seriously and assess whether, and under what conditions, inflation expectations behave like an amplification channel for energy-price shocks.

You will pursue one of two empirical strategies. In a cross-area extension, you will replicate the UK-style analysis for another jurisdiction (the US, the euro area, or selected euro-area countries) and compare dynamic responses using a small VAR or local projections, combining an energy-price shock measure with inflation, real activity, and inflation expectations drawn from standard survey and/or market-based sources. Alternatively, you will study whether the expectations channel varies over time—for example, whether it becomes more important when household price attention is plausibly higher during crisis episodes—by estimating a time-varying parameter VAR, a Makov-Switching VAR, and/or a stochastic-volatility VAR and contrasting pre-episode versus episode dynamics.

References

Transmission Channel Analysis in Dynamic Models Enrico Wegner, Lenard Lieb, Stephan Smeekes and Ines Wilms; 2025; doi:10.48550/arXiv.2405.18987

Structural Vector Autoregressive Analysis Lutz Kilian and Helmut Lütkepohl; 2017; Cambridge University Press

New Introduction to Multiple Time Series Analysis Helmut Lütkepohl; 2005; Springer

2.7 Sampling Deferred Acceptance algorithm with tie-breaking in school choice

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Discipline: Econometrics, Mathematical Economics, Operations Research

Description

In many cities around the world, a centralized algorithm decides how the scarce high-school seats are assigned to prospective students. The most commonly used algorithm for this problem is the Deferred Acceptance (DA) algorithm (see, e.g., Roth, 2008). The DA algorithm requires both the students and the schools to have a strict preference ordering over the other side of the market. While the students often have such strictly ordered preference lists, schools often have *weak* preferences over the students. These weak preferences can be caused by coarse criteria, such as whether the applying student has a sibling at that school, or based on distance ranges from the applying students to the schools.

In order to obtain strict preferences and run the DA algorithm, it is common practice to break ties in the weak preferences in a random manner (e.g., Erdil & Ergin, 2008), and then run DA for this modified instance. As such, the mechanism becomes a probabilistic rule, where the assignment probabilities for each student-school pair are specified. However, it is known that computing the expected assignment probabilities of DA with random tie-breaking is $\#P$-hard, which implies that it is intractable in practice (Aziz et al, 2013).

In this thesis, you will investigate the accuracy of sampling methods to estimate DA with random tie-breaking. This can be done empirically by sampling increasing numbers of random tie-breakings (i.e., random orderings of the students), and by computing confidence intervals on these estimates, through, for example, methods such as bootstrapping.

References

Aziz, H., Brandt, F., & Brill, M. (2013). The computational complexity of random serial dictatorship. Economics Letters, 121(3), 341-345.

Erdil, A., & Ergin, H. (2008). What's the matter with tie-breaking? Improving efficiency in school choice. American Economic Review, 98(3), 669-689.

Roth, A. E. (2008). Deferred acceptance algorithms: History, theory, practice, and open questions. international Journal of game Theory, 36(3-4), 537-569.

2.8 Deep Learning Methods for Time Series Analysis: Neural Network Particle Filters for Time Series Data

Supervisor: Nalan Basturk
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Discipline: Econometrics
Required Courses: EBC2120 Econometric Methods II, EBS2072 Introduction to Software in Econometrics

Description

In this thesis you will study a recent neural network–based particle filtering method for estimating time-varying probability densities in time series data. Estimating full predictive distributions, rather than only point forecasts, is central in econometrics and forecasting, especially in the presence of non-Gaussianity, nonlinear dynamics, and time-varying uncertainty.

The thesis builds on a recent contribution by Peerlings et al. (2026), which combines neural network density estimation with particle filtering techniques to obtain flexible, nonparametric density forecasts. The method allows for rich distributional features such as skewness, heavy tails, and multimodality.

Your main task will be to implement this methodology and apply it to a new empirical dataset (for example financial returns, electricity prices, macroeconomic indicators, or environmental time series). You will evaluate the performance of the neural network particle filter relative to benchmark methods such as kernel density estimation, parametric distributional models, or classical state-space approaches.

The empirical comparison will be conducted using appropriate scoring rules and goodness-of-fit measures, such as log-likelihood or out-of-sample predictive scores. Particular attention will be paid to the model's ability to capture tail risk and time-varying uncertainty.

This project combines econometric theory, computational implementation, and empirical evaluation. It is suitable for students with strong quantitative skills and an interest in modern machine learning methods in econometrics.

References

Neural Network Particle Filter for Time Series Data Peerlings, D. E., van den Brakel, J. A., Baştürk, N., & Puts, M. J., 2026, Journal of Forecasting, forthcoming.

Multivariate density estimation by neural networks Peerlings, D. E., van den Brakel, J. A., Baştürk, N., & Puts, M. J., 2022, IEEE Transactions on Neural Networks and Learning Systems, 35(2), 2436-2447.

2.9 A Simulation-Based Comparison of N-BEATS and Classical Econometric Models for Multivariate Time Series Forecasting

Supervisor: Rui Jorge Almeida
Email: rj.almeida@maastrichtuniversity.nl
Discipline: Econometrics
Required Courses: EBC2120 Econometric Methods II

Description

N-BEATS is a deep learning model designed to replicate the forecasting mechanisms of traditional statistical models using stacks of fully connected layers. This model possesses several desirable properties, including interpretability, computational efficiency, and applicability across various domains. A key advancement in its development is the extension to multivariate time series and the inclusion of covariates, as implemented in the Darts library.

The objective of this thesis is to conduct a comprehensive simulation study comparing the univariate and multivariate versions of N-BEATS with well-established econometric models, such as Vector Autoregression (VAR) and Seasonal Autoregressive Integrated Moving Average (SARIMA). The comparison will not only focus on traditional performance metrics (such as accuracy and computational efficiency) but also delve into the interpretability of the models.

Specifically, the study will examine whether both modeling approaches capture and convey the same underlying signals in the data. It will analyze the decomposition structures inherent in econometric models, such as the explicit representation of trend, seasonality, and exogenous influences, and compare them to the learned representations within N-BEATS. Additionally, the ability of each model to provide insights into variable relationships, causality, and the influence of external factors will be assessed.

References

N-BEATS: Neural basis expansion analysis for interpretable time series forecasting. Oreshkin, B. N., Carpov, D., Chapados, N., & Bengio, Y. (2019). arXiv preprint arXiv:1905.10437.

Darts: User-friendly modern machine learning for time series. Herzen, J., Lässig, F., Piazzetta, S. G., Neuer, T., Tafti, L., Raille, G., & Grosch, G. (2022). Journal of Machine Learning Research, 23(124), 1-6.

2.10 Replication and Cross-Market Validation of Deep Learning-Based Price Trend Extractio

Supervisor: Rui Jorge Almeida
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Discipline: Econometrics
Required Courses: EBC2120 Econometric Methods II

Description

The objective of this study is to replicate and extend the key findings of the paper "(Re-)Imag(in)ing Price Trends", applying their methodology to different types of commodities. The original work leverages advanced machine learning techniques to extract and analyze price trends, demonstrating the power of deep learning in financial forecasting. By applying similar methods to a diverse set of commodities, this research aims to evaluate the robustness and adaptability of the approach in different market contexts.

References

(Re-) Imag (in) ing price trends. Jiang, J., Kelly, B., & Xiu, D. (2023). The Journal of Finance, 78(6), 3193-3249.

Price movements in speculative markets: Trends or random walks. Alexander, S. S. (1961). Industrial Management Review (pre-1986), 2(2), 7.

2.11 Comparison of tuning parameter selection methods for sparse regressions

Supervisor: Ines Wilms
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Discipline: Econometrics
Required Courses: EBC2120 Econometric Methods II, EBS2072 Introduction to Software in Econometrics

Description

High-dimensional regression models have become a standard choice for modeling complex relationships involving a large number of covariates. However, when the number of covariates becomes large relative to the sample size, they become overparametrized. One option to address this curse of dimensionality is to resort to the lasso (Tibshirani, 1996) which permits estimation of the regression parameters in such high-dimensional settings. The lasso returns a sparse solution where many of the regression coefficients are put to zero.

A key component of the lasso is the selection of its tuning parameter that regulates the degree of sparsity of the solution. To select this tuning parameter, one can resort to cross-validation,

or information criteria such as AIC or BIC. Recently, Sadhukhan et al. (2025) proposed another data-driven approach to select the tuning parameter, which you will explore in this thesis.

You will explore high-dimensional regression models estimated by the lasso and compare different techniques for selecting the tuning parameter through carefully set-up Monte Carlo simulation experiments. In particular, you will

- study high-dimensional regression models and the lasso (Tibshirani, 1996).
- explore the literature on sparse regression models and tuning parameter selection for the lasso.
- learn how to carefully set-up Monte Carlo simulation studies, and you will compare different approaches to select the tuning parameters through a variety of Monte Carlo experiments.

Importantly, good programming skills in R are required to successfully conduct this bachelor thesis topics. You need to be able to work independently in R since you will need to code up all your Monte Carlo simulation experiments!

References

Sadhukhan, T., Wilms, I., Smeekes, S., Basu, S. (2025). Autotune: fast, accurate, and automatic tuning parameter selection for LASSO. arXiv preprint arXiv:2512.11139.

Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 58(1), 267-288.

2.12 Unraveling GNSS Errors From Geomagnetic Indices

Supervisor: Martin Schumann and Ines Wilms
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Discipline: Econometrics
Required Courses: EBC2120 Econometric Methods II, EBS2072 Introduction to Software in Econometrics

Description

Global Navigation Satellite Systems, GNSS, such as GPS and Galileo are essential for services all over the world, with both civil and military applications, but the quality of these signals is not guaranteed. Before reaching the ground, the GNSS signals must pass through the Earth's ionosphere. The structure of the ionosphere is dynamic and often responds strongly to external drivers, including the solar and geomagnetic activity. With disturbed ionosphere, the signals propagating through it may be distorted by its irregular structure. Especially in equatorial regions and polar regions, these effects are more pronounced and have thus been well studied. However, after strong geomagnetic storms, GPS accuracy has been observed to decrease even in mid-latitude regions (including the Netherlands).

In this thesis project, the student is expected to investigate the relationship between various indicators of geomagnetic activity, such as the Kp geomagnetic index, and the GNSS error at multiple locations in Europe obtain from the European Geostationary Navigation Overlay Service

(GNSS) to determine to what extent the former can be used to predict the latter, and whether this is even possible in the first place. Some of the challenges here include time series with different temporal resolutions and modelling multiple stations which may respond differently to the same phenomena.

Importantly, good programming skills (for instance in R or Python) are required to successfully conduct this bachelor thesis topics. You need to be able to work independently in such software languages to pre-process the data and conduct the analyses! You also need to have a solid interest in physics.

References

Bakota, M., Jelaska, I., Kos, S., Brcic, D. (2025). Statistical Approach to Research on the Relationship Between Kp/Dst Geomagnetic Indices and Total GPS Position Error. Remote sensing, 17(14), 2374.

Veettil, S. V., Aquino, M., Spogli, L., Cesaroni, C. (2018). A statistical approach to estimate Global Navigation Satellite Systems (GNSS) receiver signal tracking performance in the presence of ionospheric scintillation. Journal of Space Weather and Space Climate, 8, A51.

Younas, W., Nishimura, Y., Liao, W., Semeter, J. L., Mrak, S., Morton, Y. J., Groves, K. M. (2025). Spatio-temporal evolution of mid-latitude GPS scintillation and position errors during the May 2024 solar storm. Journal of Geophysical Research: Space Physics, 130(6), e2025JA033839.

2.13 A Statistical Analysis of Geostationary and Low-Earth Orbit Environments

Supervisor: Martin Schumann and Ines Wilms
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Discipline: Econometrics
Required Courses: EBC2120 Econometric Methods II, EBS2072 Introduction to Software in Econometrics

Description

Satellites are continuously exposed to energetic particles originating from the Sun and the Earth. Dutch satellites currently operate mainly in low-Earth orbit (LEO), where the particle environment can vary strongly in response to solar and geomagnetic activity. However, real-time measurements of this environment are limited. At present, satellite operators rely on real-time particle measurements from satellites in geostationary orbit (GEO) to assess space weather conditions and to advise LEO satellite operators. While this approach is practical, it is known that the particle environment in GEO does not directly correspond to conditions in LEO, meaning that the advice given may be inaccurate. Fortunately, there exists a large historical dataset of particle measurements from LEO satellites covering roughly the last 20 years, although these data are not available in real time.

In this thesis project, the student is expected to investigate the relationship between particle measurements in GEO and LEO satellites. By statistically comparing historical measurements from both orbital regions, the goal is to determine whether real-time GEO observations can be

used to better estimate particle conditions in LEO, and whether systematic corrections to current operational advice are possible (Buitrago-Leiva et al., 2024). Key challenges include working with high-frequency time series at different temporal resolutions, as also done in van Ravenswaaij et al. (2025).

Importantly, good programming skills (for instance in R or Python) are required to successfully conduct this bachelor thesis topics. You need to be able to work independently in such software languages to pre-process the data and conduct the analyses! You also need to have a solid interest in physics.

References

van Ravenswaaij, J., Wilms, I., Ricardo I., Brchnelova, M. (2025). Performance of radio-based detection to operationally monitor M5+ class solar flares, Working paper.

Buitrago-Leiva, J. N., El Khayati Ramouz, M., Camps, A., Ruiz-de-Azua, J. A. (2024). Statistical analysis of LEO and GEO satellite anomalies and space radiation. Aerospace, 11(11), 924.

2.14 Temporal Aggregation for Microclimate Temperature Forecasting

Supervisor: Marie Corillon
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Discipline: Econometrics
Required Courses: EBC2120 Econometric Methods II, EBS2072 Introduction to Software in Econometrics

Description

High-frequency time series forecasting often requires many lags, which quickly creates high-dimensional models that are noisy, hard to estimate, and difficult to interpret. StarTime is a penalized regression method that tackles this by performing temporal aggregation of lags in a data-driven way: for example, it may find that several daily lags should enter the model as a single weekly effect, and several weekly effects as a monthly effect, while still allowing important short-run lags to remain separate when needed. This reduces dimensionality, improves the signal-to-noise ratio, and yields interpretable multi-horizon dynamics (in the spirit of the HAR-type idea by Corsi (2009), but without fixing the horizons in advance).

In this bachelor thesis, you will apply StarTime to forecast temperatures across different locations using a microclimate dataset with very high-frequency (30-minute) in-situ measurements. Microclimate refers to the temperature conditions close to the ground or within sheltered habitats, which can differ substantially from standard "open-air" weather-station measurements and is often more relevant for ecosystems. You will compare StarTime's forecasting performance to standard high-dimensional penalized benchmarks (which mainly control complexity by shrinking many coefficients) and assess when data-driven temporal aggregation helps most for temperature prediction.

Good programming skills in R are required to successfully conduct this bachelor thesis topics. You need to be able to work independently in R since you need to pre-process data and conduct the analyses.

References

Corillon, M., Wilms, I., Smeekes, S. (2026). Sparse Tree-Based Aggregation for Time Series Regressions. [Working Paper].

Corsi, F. (2009). A Simple Approximate Long-Memory Model of Realized Volatility. *Journal of Financial Econometrics*, 7(2), 174-196.

Lembrechts, J. J., Aalto, J., Ashcroft, M. B., De Frenne, P., Kopecký, M., Lenoir, J., ... & Rocha, A. (2020). SoilTemp: A global database of near-surface temperature. *Global change biology*, 26(11), 6616-6629.

Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 58(1), 267-288.

2.15 The Business Cycle Paradox: Balancing Statistical Rigor and Economic Reality

Supervisor: Tobias Hartl
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Discipline: Econometrics

Description

Measuring the business cycle, that is the cyclical fluctuations of economic activity around its long-term trend, remains one of the most contentious tasks in macroeconometrics. The central "puzzle" lies in a persistent gap between statistical validity of trend-cycle models and economic plausibility of the resulting decomposition.

On one hand, many economists rely on the Hodrick-Prescott (HP) filter; while it produces smooth cycles that align with our intuitive understanding of recessions, it is often criticized for being a statistically misspecified tool that can generate spurious cycles even in random data. On the other hand, rigorous state-space models and unobserved components models often yield trend-cycle decompositions that are statistically sound but economically baffling. For instance, Morley, Nelson, and Zivot (2003) demonstrated that when allowing for correlation between trend and cycle innovations, the estimated business cycle becomes almost non-existent or fails to match historical recession dates.

In this thesis, you will explore this decomposition dilemma. The core of the project involves: 1. Replicating some of the prominent business cycle estimates found in the literature (e.g., comparing the results of Morley, Nelson, and Zivot (2003) with the structural break approach of Perron and Wada(2009)). 2. Investigating how specific model assumptions, such as the correlation between innovations or the presence of structural breaks, alter our perception of economic health. 3. Assess whether the resulting trend-cycle decompositions satisfy the assumptions imposed on the models.

References

Why are the Beveridge-Nelson and Unobserved Components Decompositions of GDP so Different? Morley, J. C., Nelson, C. R., & Zivot, E. (2003), Review of Economics and Statistics.

Let's take a break: Trends and cycles in US real GDP. Perron, P., & Wada, T. (2009), Journal of Monetary Economics.

Why You Should Never Use the Hodrick-Prescott Filter. Hamilton, J. D. (2018), Review of Economics and Statistics.

2.16 The Great Filter Debate: Hamilton vs. Hodrick and Prescott

Supervisor: Tobias Hartl
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Discipline: Econometrics

Description

For years, the Hodrick-Prescott (HP) filter has been the standard for decomposing time series into trend and cycle - despite a lot of criticism from the field of statistics. Recently, Hamilton (2018) argued the HP filter should be abandoned because it introduces spurious cycles among others, and proposed a simple, regression-based alternative. Philips and Shi (2021) argued the HP filter can be fixed by a boosting algorithm: it repeatedly applies the HP filter until a convergence criterion is hit.

The core of this paper is to conduct a robust comparison of these three competing trend-cycle models. You will 1. Select macroeconomic time series (e.g., Real GDP, Investment, or Consumption) 2. Implement the classic HP filter, Hamilton's regression-based approach, and the iterative Boosted HP (bHP) filter. 3. Compare how these filters handle volatility, such as the COVID-19 pandemic shock, cyclical asymmetries, and other anomalies, and assess which method provides the most reliable estimate of the output gap. 4. Use Monte Carlo simulations to verify your empirical results 5. Investigate whether the boosting process truly solves the statistical misspecification issues highlighted by Hamilton, or if the regression approach remains superior.

References

Why You Should Never Use the Hodrick-Prescott Filter. Hamilton, J. D., (2018), Review of Economics and Statistics.

Boosting: Why you can use the HP filter. Phillips, P. C. B., & Shi, Z., (2021), International Economic Review

Reliable Real-Time Output Gap Estimates Based on a Modified Hamilton Filter. Quast, J., & Wolters, M. H. (2022), Journal of Business & Economic Statistics.

2.17 Have financial markets become more volatile? A factor-based investigation

Supervisor: Tobias Hartl
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Discipline: Econometrics

Description

In this thesis, you will investigate the dynamics of systemic risk and market instability. Some events impact the volatilities of virtually all assets traded on financial markets; prominent examples include the 2008 Global Financial Crisis, the COVID-19 pandemic market shock, and significant geopolitical conflicts. Because these shocks permeate the entire financial system, market volatility can be conceptualized as a latent common factor that drives the individual variances of diverse assets simultaneously.

The core of this project is to measure and analyze common volatility on financial markets. The statistical building block will be factor models, a model class that allows to compress common variation in the cross-section into a small set of common factors. These factors can be thought as the underlying market sentiment that drives a set of volatility measures, e.g. price variation of all stocks contained in a large index.

Your research project will involve: 1. Collecting daily data for individual volatility proxies (such as the price spreads or realized volatilities of a broad cross-section of stocks). 2. Implementing a factor model to extract the common volatility factor. 3. Analyzing the dynamic behavior of common volatility and comparing its peaks to significant historical and economic events to determine if the "pulse" of the market has truly become more erratic.

References

Dynamic Factor Models, Factor-Augmented Vector Autoregressions, and Structural Vector Autoregressions in Macroeconomics. Stock, J. H. & Watson, M.W. (2016). Handbook of Macroeconomics, vol. 2

Forecasting using principal components from a large number of predictors. Stock, J. H., & Watson, M. W. (2002). Journal of the American Statistical Association.

Generalized dynamic factor models and volatilities: recovering the market volatility shocks. Barigozzi, M., & Hallin, M., (2016), The Econometrics Journal.

2.18 Forecasting Cryptocurrencies Volatility using GARCH models

Supervisor: Daniel Velasquez-Gaviria
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Discipline: Econometrics
Required Courses: EBC2120 Econometric Methods II, EBS2072 Introduction to Software in Econometrics

Description

Cryptocurrencies display significantly higher volatility compared to traditional assets, making them riskier investments. Accurate short-term volatility forecasts can help market participants manage risk, optimize buy/sell timing, and hedge against extreme losses. This study proposes identifying the most suitable GARCH (Generalized Autoregressive Conditional Heteroskedasticity) model specifications for capturing the conditional volatility of both major cryptocurrencies (e.g., Bitcoin, Ethereum) and emerging "memecoins." It will also explore which probability distributions (e.g., normal, Student's t) best fit daily returns, ultimately creating decision criteria that distinguish

periods of low versus high volatility. Familiarity with time-series analysis and programming in R or Python is recommended to carry out this research.

Research Questions: (1) Which GARCH model variant (e.g., GARCH, EGARCH, GJR-GARCH) most accurately captures the conditional volatility of leading cryptocurrencies and memecoins? (2) Which probability distribution provides the best fit and best forecast for daily cryptocurrency volatility? (3) How can the identified models and distributions inform practical decision criteria for managing risk and optimizing market entry/exit timing under varying volatility regimes?

References

A conditionally heteroskedastic time series model for speculative prices and rates of return Bollerslev, T., 1987, The Review of Economics and Statistics, 542–547.

Testing expected shortfall: an application to emerging market stock indices Cardona, Emilio; Mora-Valencia, Andrés; Velásquez-Gaviria, Daniel, 2019, Risk Management, 21, 153–182.

Forecasting cryptocurrency volatility Catania, Leopoldo; Grassi, Stefano, 2022, International Journal of Forecasting, 38(3), 878–894.

A comparison of the risk quantification in traditional and renewable energy markets Velásquez-Gaviria, D.; Mora-Valencia, A.; Perote, J., 2020, Energies, 13(11), 2805.

2.19 Comparative Analysis of Market Risk in AI-Driven Tech Stocks vs. Large Value Stocks Using Value at Risk

Supervisor: Daniel Velasquez-Gaviria
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Discipline: Econometrics
Required Courses: EBC2120 Econometric Methods II, EBS2072 Introduction to Software in Econometrics

Description

The rapid rise of artificial intelligence has fueled a significant rally in the prices of technology stocks—particularly those involved in AI—from 2021 to the present. These tech firms are typically classified as growth companies, where their current valuations rely more on future cash flow expectations than on intrinsic value. While this presents substantial profit opportunities, it also increases the market risk, especially compared to traditional value stocks that demonstrate more stable, intrinsic-based valuations.

This thesis aims to compare the market risk faced by investors in AI-focused U.S. tech stocks with that of large value stocks, using Value at Risk (VaR) as the primary risk metric. The study will employ both historical simulation and ARMA-GARCH models with various probability distributions to estimate VaR, followed by backtesting procedures to determine which approach provides the most reliable protection against extreme losses.

Research Questions

(1) How does the VaR of AI-driven tech stocks compare to that of large value stocks?

(2) Which estimation technique (historical simulation vs. ARMA-GARCH models with different probability distributions) most accurately captures extreme downside risk?

(3) What insights can backtesting provide regarding the effectiveness of these VaR methods in mitigating potential extreme losses in volatile markets?

References

A conditionally heteroskedastic time series model for speculative prices and rates of return Bollerslev, T., 1987, The Review of Economics and Statistics, 542–547.

Testing expected shortfall: an application to emerging market stock indices Cardona, Emilio; Mora-Valencia, Andrés; Velásquez-Gaviria, Daniel, 2019, Risk Management, 21, 153–182.

A comparison of the risk quantification in traditional and renewable energy markets Velásquez-Gaviria, D.; Mora-Valencia, A.; Perote, J., 2020, Energies, 13(11), 2805.

2.20 Data Analysis and Optimisation of the Maastricht Digital Twin

Supervisor: Nalan Basturk, Andre Berger
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Discipline: Econometrics
Required Courses: EBC2120 Econometric Methods II

Description

This thesis examines whether selected urban policies and major events in Maastricht had short-term or persistent effects on visitor numbers, dwell time, and spatial distribution in the city centre. Using hourly Digital Twin data, the student will apply time-series and causal inference techniques to analyze the number of visitors in Maastricht. The thesis will be conducted and supervised in collaboration with Argaleo, with Centrummanagement Maastricht as the host institution.

The purpose of this study is to analyze the dynamics of visitor counts in Maastricht using a recently available, high frequency dataset from Digital Twin and state of the art econometric methods. More specifically, we want to understand the causal impact of specific policy interventions or major events in Maastricht on urban visitor dynamics using high-frequency panel data from the Maastricht Digital Twin. The student will select one intervention—such as the Zero Emission Zone (ZES), a major redevelopment (e.g., Fronten parking facility), or a large-scale event (e.g., André Rieu concerts or Carnival)—and employ econometric techniques will be used to determine whether observed changes in visitor counts, dwell time, or district-level crowding were transitory or structurally persistent. The empirical analysis will rely on hourly data for the city centre and its districts (Binnenstad, Wyck, Jekerkwartier, Sphinxkwartier, and Stokstraatkwartier), potentially controlling for confounding factors such as weather, seasonality, holidays, and day-of-week effects. The core contribution is to understand general trends and seasonal patterns in visitor counts, and to understand the immediate and long-run effects.

References

Wooldridge, J. M. (2019). *Introductory Econometrics: A Modern Approach* (7th ed.). Cengage Learning.

Forecasting city arrivals with Google Analytics Gunter, U., & Onder, I. (2016). Annals of Tourism Research, 61, 199-212.

2.21 Does Inequality hurt Growth?

Supervisor: Martin Schumann & Ties Bos
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Discipline: Econometrics
Required Courses: EBC2120 Econometric Methods II

Description

In this thesis, you will investigate the macroeconomic relationship between income inequality and economic growth. For decades, economists debated whether redistribution hurts efficiency (the "Okun Gap"). In a landmark study, Berg et al. (2018) used dynamic panel data models with GMM to argue that lower net inequality is robustly correlated with faster and more durable growth.

However, the GMM estimator used in this literature is very sensitive to "instrument proliferation" and weak identification. Your task is to critically assess whether the "Inequality hurts Growth" result holds up to modern econometric standards and recent data.

The data set used in the original study (SWIID for inequality, Penn World Tables for growth) are open-access and continuously updated, which allows you to include more recent waves of data in your analysis.

The core task of your thesis is to replicate the baseline dynamic panel model from Berg et al. (2018) for the period 1960–2010. Moreover, you should update the data and check if the results still hold.

After the core task, you will be checking the econometric approach of the original study for robustness. One potential weak point is that the authors use many weak instruments. Instead of using all instruments, you could check for methods that use a subset of the "best" instruments. The original study also assumes that the effect of inequality on growth is the same for developing and developed nations. You could "zoom in" to analyze the heterogeneity of the effect.

References

Redistribution, inequality, and growth: new evidence Berg, A., Ostry, J. D., Tsangarides, C. G., & Yakhshilikov, Y., 2018, Journal of Economic Growth.

How to do xtabond2: An introduction to difference and system GMM in Stata Roodman, D., 2009, Stata Journal.

The Standardized World Income Inequality Database Solt, F., 2020, Social Science Quarterly.

2.22 Gun Laws and Crime: Re-evaluating 'Stand Your Ground' with Heterogeneous DiD

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Discipline: Econometrics
Required Courses: EBC2120 Econometric Methods II

Description

"Stand Your Ground" laws extend the self-defense rights of individuals, removing the duty to retreat before using lethal force. In a seminal paper, Cheng and Hoekstra (2013) used a Two-Way Fixed Effects (TWFE) Difference-in-Differences model to estimate the impact of these laws on homicide rates. They found that the laws did not deter crime but led to a significant increase in homicides. The laws were implemented in different states at different times (staggered adoption).

Recent econometric literature (e.g., Goodman-Bacon, 2021; Callaway and Sant'Anna, 2021) has shown that when treatment timing is staggered and treatment effects vary over time (e.g., the effect of the law grows stronger the longer it is in place), the standard TWFE estimator is biased. It effectively uses "early treated" units as controls for "late treated" units, which can flip the sign of the estimated coefficient even if the true effect is positive.

In this thesis, you will test if the findings of Cheng and Hoekstra (2013) still hold with modern methods that allow for treatment effect heterogeneity.

References

Callaway, B., & Sant'Anna, P. H. (2021). *Difference-in-differences with multiple time periods. Journal of Econometrics*, 225(2), 200-230.

Cheng, C., & Hoekstra, M. (2013). *Does strengthening self-defense law deter crime or escalate violence? Evidence from expansions to castle doctrine. Journal of Human Resources*, 48(3), 821-854.

Goodman-Bacon, A. (2021). *Difference-in-differences with variation in treatment timing. Journal of Econometrics*, 225(2), 254-277.

2.23 The Economic Costs of Brexit: A Synthetic Difference-in-Differences Approach

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Discipline: Econometrics
Required Courses: EBC2120 Econometric Methods II

Description

In June 2016, the United Kingdom voted to leave the European Union. Almost immediately, economists began debating the economic consequences of this decision. In a highly influential paper, Born et al. (2019) utilized the Synthetic Control Method (SCM) to construct a synthetic counterfactual UK. They found that the vote caused a significant output loss compared to the synthetic counterfactual, even before the actual exit took place.

However, the original study only covered data up to 2018. Furthermore, standard SCM relies heavily on the quality of the pre-treatment fit and does not account for unit-specific fixed effects. Arkhangelsky et al. (2021) recently proposed the Synthetic Difference-in-Differences (SDID) estimator, which combines the re-weighting benefits of SCM with the fixed-effects robustness of DiD, often yielding more precise estimates with lower bias.

In this thesis, you will update the data and investigate if the original findings still hold. Moreover, you will compare the results obtained from a synthetic control (as derived in the original study) with the results from synthetic Difference-in-Differences.

References

Arkhangelsky, D., Athey, S., Hirshberg, D. A., Imbens, G. W., & Wager, S. (2021). *Synthetic difference-in-differences. American Economic Review*, 111(12), 4088-4118.

Born, B., Müller, G. J., Schularick, M., & Sedlá?ek, P. (2019). *The costs of economic nationalism: Evidence from the Brexit experiment. The Economic Journal*, 129(623), 2722-2744.

Abadie, A. (2021). *Using synthetic controls: Feasibility, data requirements, and methodological aspects. Journal of Economic Literature*, 59(2), 391-425.

3 Mathematical Economics

3.1 Sharing money with a status quo

Supervisors: Bas Dietzenbacher en Pauline Hanssen
Email: b.dietzenbacher@maastrichtuniversity.nl and pauline.hanssen@maastrichtuniversity.nl
Discipline: Mathematical Economics

Description

An amount of money needs to be divided among a group of agents. These agents have status quo payoffs that reflect how money is allocated at this moment, but the amount of money to be divided may change over time. In particular, the total amount to divide may exceed the aggregate status quo or fall short. How should the money be divided among the agents? You will define several sharing rules and compare them on the basis of their properties.

3.2 Division of time among agents

Supervisors: Bas Dietzenbacher en Pauline Hanssen
Email: b.dietzenbacher@maastrichtuniversity.nl and pauline.hanssen@maastrichtuniversity.nl
Discipline: Mathematical Economics

Description

A group of agents is assigned to a project that requires a certain amount of time. Each agent has a most preferred amount of time to spend on the project, and preferences decrease in both directions (less/more time to spend on the project is less preferred). The total preferred amount of time to spend on the project may exceed the requirement or fall short. How should the project time be divided among the agents? You will define several division rules and compare them on the basis of their properties.

3.3 Allocating extra prize money in competitions

Supervisor: Bas Dietzenbacher
Email: b.dietzenbacher@maastrichtuniversity.nl
Discipline: Mathematical Economics

Description

Many sport competitions result in a final ranking. Given the ranking of competitors, how should the prize endowment be allocated? There are several existing families of allocation rules solving this issue. All these rules compromise between the equal division rule and the winner-takes-all rule. Now suppose there is additional prize money coming from a new sponsor or a new broadcasting channel. How should this additional prize money be allocated among the competitors? We formulate properties for that issue and study which rules satisfy these properties. Our results may help organizers to select the most suitable prize allocation rule for rank-order competitions.

References

Fair and consistent prize allocation in competitions Dietzenbacher and Kondratev, 2023, Management Science.

3.4 Supra-efficient allocation and budget balancedness

Supervisor: Dries Vermeulen
Email: d.vermeulen@maastrichtuniversity.nl
Discipline: Mathematical Economics
Required Courses: EBC2091 Market Design

Description

In this thesis we focus on allocation problems in which multiple heterogeneous indivisible items are assigned to a group of interested buyers. You will investigate whether allocation schemes that allocate more items than total utility maximization allows can be implemented via a budget-balanced auction.

We start with a simple model in which two items are allocated to two potential buyers. Taking those results as a basis, the project can be extended in several directions, varying the number of items and buyers, as well as the characteristics of the items.

References

An efficient ascending-bid auction for multiple objects Lawrence M. Ausubel, 2004, American Economic Review.

Multi item auctions Demange, Gale, Sotomayor, 1986, The Journal of Political Economy.

3.5 Competition between Fishing Companies: A Game-Theoretic Analysis

Supervisor: Andrés Perea
Email: a.perea@maastrichtuniversity.nl
Discipline: Mathematical Economics

Description

Consider a situation where two fishing companies fish on the same lake during many years. Every year, the two companies must decide how much fish to catch. Catching more has two opposite effects: It increases the immediate profit, but it also decreases the amount of fish that will be left next year.
The purpose of this bachelor thesis is three-fold: First, try to create a finite dynamic game with finite horizon that incorporates the effects described above. Subsequently, analyze the game by using the *double utility procedure* proposed in Perea (2025). Finish with a comparitive statics analysis, in which you investigate how the optimal behavior of the companies depends on (a) the number of periods, and (b) how quickly, or slowly, the fish population recovers after heavy fishing.

References

Pure backward induction reasoning in dynamic games Perea, A. (2025), Epicenter Working Paper No. 34.

3.6 Largest unique integer game

Supervisor: Marc Schröder and Anh Trieu
Email: m.schroder@maastrichtuniversity.nl/anh.trieu@maastrichtuniversity.nl
Discipline: Mathematical Economics

Description

We thoroughly study the largest unique integer game as introduced in Game Theory and Economics. In this game the player with the largest integer that is unique wins. Given a fixed set of numbers, the questions we would like to address are as follows:

1. Do pure Nash equilibria always exist?
2. Do pure Nash equilibria always have a winner, or does this depend on the number of players?
3. How does the probability of having a winner change dependent on the number of players in the symmetric Nash equilibrium?

References

Testing game theory in the field: Swedish LUPI lottery games Östling et al. (2011).

Comments on 'Reverse auction: the lowest unique positive integer game' Flitney (2008).

3.7 Contests with consolation prizes

Supervisor: Marc Schröder
Email: m.schroder@maastrichtuniversity.nl
Discipline: Mathematical Economics

Description

Many contests do not only award a prize to the winner, but also to the player that ends last. We would like to study what the incentive of such a "loser trophy" is. Contests are often modeled via an all-pay auction: every player pays their effort and the player with the highest effort wins. We would like to study the impact of having a small prize for the loser on the efforts of the participants in an all-pay auction.

References

Politically contestable rents and transfers Hillman and Riley (1989).

Optimal seedings in elimination tournaments Groh et al. (2012).

3.8 Hotelling with congestion effects

Supervisor: Marc Schröder
Email: m.schroder@maastrichtuniversity.nl
Discipline: Mathematical Economics

Description

Hotelling's location game is one of the simplest, but most famous location game. Given an interval, the question is where to locate your firm. In this variant of the model, we consider consumers that care about travel distances as well as waiting costs. In the game with two firms, a pure Nash equilibrium only exists if both firms have the same waiting cost coefficient. In this thesis, you will study mixed Nash equilibria of this game based on results of a recent doctoral thesis by Joep van Sloun. The idea is to use the two rationalizable locations to find mixed Nash equilibria first in a finite game and later in the infinite game.

References

Hotelling's location model with negative network externalities Peters et al. (2018).

Rationalizable behavior in the hotelling model with waiting costs Van Sloun (2025).

3.9 Choice via AI

Supervisor: (Jan) Christopher
Email: j.kops@maastrichtuniversity.nl
Discipline: Mathematical Economics

Description

Suppose that a decision maker (DM) seeks advice from an AI Agent before choosing from a menu of available choices. Based on the recommendations that she receives, she is interested in answering three questions. First, does the AI make rational recommendations? Second, are these recommendations aligned with her own preferences? Third, how can we guarantee that the recommendations remain grounded within the original feasible set?

The aim of this thesis is to characterize the AI's recommendation "behavior". That is, we seek conditions on the AI's recommendations and (mis)interpretations that allow us to address the questions above.

This is a project in theoretical economics. We are neither programming an AI, nor are we really working with AI. Instead, we adopt the perspective of axiomatic choice theory, asking abstract questions of the sort: what axioms an AI should or must satisfy and how we can rationalize the AI's recommendations.

References

Manzini, Paola and Mariotti, Marco (2007). *Sequentially Rationalizable Choice*. American Economic Review, 97(5):1824–1839.

Kops, Christopher, Elias Tsakas (2026) *Choice via AI*, working paper

3.10 Identifying Behavioral Types

Supervisor: (Jan) Christopher
Email: j.kops@maastrichtuniversity.nl
Discipline: Mathematical Economics
Required Courses: sufficient interest in math, especially in Linear Algebra

Description

Latent factors such as preferences, attention, and cognitive capacity influence choice behavior. These factors are typically heterogeneously distributed—be it across individuals in a population or within a single decision maker observed repeatedly over time. This heterogeneity then gives rise to various *behavioral types*. Whenever the individual choices of these types remain unobservable—for example because of privacy protection, anonymization, or the sheer scale of data—they may still produce observable choice data on the aggregate level. This raises a fundamental question: when is it possible to infer the distribution of behavioral types, or their type-specific choice patterns, from aggregate choice data alone?

The aim of this thesis is to identify behavioral types from choice data. To this end, we study a general framework of stochastic choice and derive conditions on the choice data that allow us to answer the identification question in the affirmative.

The project is rather theoretical in nature, and requires sufficient interest in mathematical reasoning and theorem proving. The model and concepts are coming from Stochastic Choice Theory, whereas most of the mathematical techniques stem from new methods in Linear Algebra.

References

Dardanoni, Valentino, Paola Manzini, Marco Mariotti, and Christopher J. Tyson (2020) *Inferring Cognitive Heterogeneity from Aggregate Choices*, Econometrica, 88(3): 1269-1296

Kops, Christopher, Paola Manzini, Marco Mariotti and Illia Pasichnichenko (2026) *Identifying Behavioral Types*, working paper

3.11 Navigating in a graph with an unreliable GPS

Supervisor: János Flesch
Email: j.flesch@maastrichtuniversity.nl
Discipline: Mathematical Economics
Required Courses: EBS2073 Introduction to Software in Operations Research

Description

The following search problem has been recently examined in the literature. There is a given undirected graph. A decision maker (DM, the searcher) is located in an initial position in this graph, and his/her goal is to reach another given position in the graph (the destination) as soon as possible. In every time period, the DM can move from his current position to any other position in the graph that is connected with his current position. The DM does not know the graph and therefore fully relies on a GPS. The GPS is however not fully trustable: in each position in the graph, with probability p the GPS works correctly and points to the next position which lies on the

shortest path to the destination, and with probability $1-p$ the GPS makes a mistake and points to another position. The DM cannot follow the GPS everywhere, because due possible mistakes of the GPS, this could lead to a loop. Instead, the DM uses a strategy which follows the GPS with probability q (trust probability) in each position, and deviates from the recommendation of the GPS with probability $1-q$.

The problem has various variants, and the goal of the thesis is to investigate one of them (the choice of the variant will be determined during the first meeting). For instance: (i) Suppose that the DM knows that the GPS is wrong in at least one (but possibly more) position. How much does this information change the optimal value of q? (ii) Suppose that the DM wants to maximize the probability of reaching the destination in T steps, for a given number T. What is the optimal q, depending on T? (iii) Suppose that the DM has two GPSs. What is the optimal q? (iv) Variants with multiple decision makers.

The project requires accurate mathematical analysis of this problem, and we use tools from Analysis, Probability Theory and Game Theory. The problem is theoretical in nature. A (smaller) part of the project might/will be to write a program (maybe in Mathematica or python), in order to understand and analyze certain strategic aspects of the model.

References will be given at a later point. It is now sufficient to look at the following paper: https://arxiv.org/pdf/2111.09093.pdf

3.12 Searching for a mobile hider in a directed bipartite graph and infinite matrix games

Supervisor: János Flesch
Email: j.flesch@maastrichtuniversity.nl
Discipline: Mathematical Economics
Required Courses: EBS2073 Introduction to Software in Operations Research

Description

This project connects two topics: (1) Infinite matrix games. In such games the matrix has infinitely many rows and infinitely many columns, and therefore the game sometimes does not have a value (and the players do not always have optimal strategies). (2) A class of search games where the analysis relies on the study and application of such infinite matrix games (but where the value does exist).

To be more precise on this class of search games: The game is played between two players, called the hider and the searcher. The game takes place in a directed graph, consisting of two partitions A and B, such that all arcs go from vertices in A to vertices in B. The (invisible) hider chooses an initial vertex (either in a vertex in A or in a vertex in B) and hides there, and at any period, if she is in a vertex in partition A then she has the option to stay or to travel along one of the outgoing arcs to a vertex in B, whereas if she is in a vertex in partition B then she cannot move any more. The searcher at each period chooses an arbitrary vertex of the graph, and if the hider is there, then the hider is found and the game ends. The hider does not observe which vertex is searched by the searcher. The searcher's goal is to minimize the expected search time, whereas the hider's goal is to maximize it. One interpretation of the game is that a rescue team searches possible locations of an injured person on two sides of a river, while considering the worse-case scenario.

Regarding the optimal search time, there are some general bounds depending on the size of A and B, but even in small examples it is not known what the optimal search time exactly is, and what the optimal strategies are. For instance, if A has two vertices and B has only one vertex, there are still many open questions.

The project requires accurate mathematical analysis of this problem, and we use tools from Analysis, Linear Algebra, Probability Theory and Game Theory. The problem is theoretical in nature. An important part of the project is to write a program (maybe Mathematica or python) to study and approximate infinite matrix games that arise in these search games.

References will be given at a later point. A different but a bit related search game can be found in Flesch J, Karagozoglu E, Perea A (2009) Optimal search for a moving target with the option to wait. Naval Research Logistics 56, 526–539. The graph is a bit different, and the hider (the object) in this case is not strategic, and just moves according to fixed probabilities.

3.13 Strategic Pricing in Cooperative Routing Games

Supervisor: Frank Phillipson
Email: f.phillipson@maastrichtuniversity.nl
Discipline: Mathematical Economics, Operations Research

Description

Consider a system with multiple logistic service providers, each provider operates its own set of import (from a depot to a customer) and export (customer $\rightarrow$ depot) trips. These trips can be paired into round-trip routes that reduce empty kilometers when combined intelligently across providers. To achieve this, providers share high-level information about their trips and a centralized or decentralized algorithm pairs import and export trips into feasible routes using an optimization method such as the Hungarian algorithm.

A key aspect of the system is that each provider assigns a price to its own trips, representing the opportunity cost (e.g., empty-kilometer cost) of performing those trips independently. These prices are private, locally computed, and determine whether a candidate route is acceptable: a route consisting of two trips is selected only if the true execution cost of the combined route does not exceed the sum of the declared trip prices. This creates a pool of feasible routes from which profit can be generated. The profit of each selected route, which arises from reduced empty driving, is distributed among the contributing providers using a rule equivalent to the Shapley value. This ensures fairness, budget balance, and stability provided that all providers declare their trip prices truthfully. However, because trip prices are self-reported and unverifiable, the pricing mechanism becomes the central strategic vulnerability in the system. A rational provider may inflate or deflate prices to manipulate which routes are selected, how much profit is generated, and how much profit it receives. Understanding these incentives is crucial for designing robust cooperative mechanisms.

Research Question: What is the optimal pricing strategy for a self-interested logistic service provider who aims to maximize its payoff by strategically misreporting trip prices in a cooperative routing system?

Research sub questions:

- How do price manipulations change the set of routes that satisfy the feasibility condition?

- Can a provider increase its Shapley-value share by inflating certain trip prices to increase the perceived marginal contribution?
- Can providers strategically price trips to block beneficial pairings for competitors or to push unfavorable routes onto them?
- When does lowering prices allow a provider to induce pairings that increase its overall yield?
- Under what conditions does the Shapley-based mechanism remain stable or incentive compatible?

References

1 *Algorithmic game theory.* Roughgarden, Tim. Communications of the ACM 53.7 (2010): 78-86.

2 *Fair division and collective welfare* Moulin, Hervé, MIT press, 2004.

3.14 Why are there four Pinky's within 500 meters in Maastricht?: Spatial competition and collusion on the Hotelling line

Supervisor: Anna Zseleva
Email: anna.zseleva@maastrichtuniversity.nl
Discipline: Mathematical Economics

Description

In the main shopping area of Maastricht, there are four Pinky waffle shops located within approximately 500 meters of each other. Why would a firm place multiple shops so close together? Could this be an equilibrium outcome?

In this thesis, you will study a Hotelling-model in which consumers are uniformly distributed along an interval and incur transportation costs when visiting a shop. We compare three different cases:

- A social planner choosing locations to maximize total welfare.
- A single firm operating multiple shops and choosing locations to maximize profit.
- Independent firms choosing locations non-cooperatively.

You will first analyze location choice with fixed prices. You will then compare equilibrium locations across cases and evaluate total transportation costs, firm profits and welfare.

If time permits, the model can be extended in one of the following directions:

- Allow firms to choose prices after locations are chosen.
- Introduce a limited walking distance, so that consumers only purchase if a shop is sufficiently close.
- Study how the number of outlets affects equilibrium clustering.

The goal is to characterize and compare equilibrium outcomes and to understand whether spatial clustering, such as the case of Pinky's in Maastricht, can be rationalized as an equilibrium phenomenon.

References

Stability in competition. Hotelling, H [1929] The Economic Journal, 39, 41–57.

3.15 Reputation and entry deterrence

Supervisor: Anna Zseleva and Dries Vermeulen
Email: anna.zseleva@maastrichtuniversity.nl and d.vermeulen@maastrichtuniversity.nl
Discipline: Mathematical Economics

Description

Some real life behavior of firms is assumed to be irrational, because they do not understand the "game" they are playing. What if the economists modeling their behavior are the ones who do not understand the underlying game?

An example of an assumed irrational behavior is when a monopoly's position is threatened by an entrant firm and after entry the monopoly uses predatory prices (see Selten (1978)). However, assuming incomplete information this behavior can be explained (Milgrom-Roberts (1982) and Kreps-Wilson (1982)).

Assume that in each time period a new firm could decide to enter the market of the monopolist. The entrant's decision is influenced by what the monopolist did in the past, because that way the entrant can update their information. This is how the monopolist builds their reputation.

We would like to work together on versions of this model. For example, imagine that the entrant could choose between entering different markets and they can observe past behavior on each market. Assuming asymmetric information, even if predatory prices are costly for the monopolist when should they apply them? Up til which time period does reputation building pay off? What do these firms do in equilibrium?

References

Reputation and imperfect information. Kreps, D. M. - Wilson, R. B. [1982] Journal of Economic Theory. Vol. 27. No 2. pp. 253-279.

Predation, reputation, and entry deterrence. Milgrom, P. - Roberts, J. [1982] Journal of Economic Theory. Vol. 27. No 2. pp. 280-312.

The chain store paradox. Selten, R. [1978] Theory and Decision. Vol. 9. No 2. pp. 127-159.

4 Operations Research

4.1 Deciding which bus lines should operate in a city.

Supervisor: Shahrezad Fahmy
Email: shahrezad.fahmy@maastrichtuniversity.nl
Discipline: Operations Research
Required Courses: EBC2121 Advanced Algorithms, EBS2073 Introduction to Software in Operations Research

Description

This topic focuses on the line planning problem in public transport. The goal of line planning is to decide which bus lines should operate in a city, based on passenger travel requests consisting of an origin and a destination.

We consider a common two-step modelling approach:

1. **Line pool generation:** A line pool containing a set of candidate bus lines is created.
2. **Line selection:** A subset of these candidate lines is selected to operate, aiming to serve passenger demand effectively.

In this thesis, you will focus on the second step of this approach. Namely, a set of passenger requests and a line pool are given, and you have to design an optimization model that selects which lines from the line pool should operate. An extension of this could be to not only select a subset of lines to operate, but also to compute a ranking indicating which lines are better and more likely to operate. For example, can we assign a probability distribution over the lines, where highly useful lines receive a higher probability and less useful lines receive a lower probability? Besides, we can also look at different coverage levels. For example, how do results change if we require full coverage (all passengers can complete their trip by bus) versus partial coverage (only a specified percentage of passengers can complete their trip by bus)?

References

Line planning in public transportation: models and methods., Schöbel, A., 2012, OR spectrum, 34(3), 491-510.

Line planning with minimal transfers., Schöbel, A., & Scholl, S., 2004, Institut für Numerische und Angewandte Mathematik der Universität Göttingen.

4.2 Evaluating the assignment of high-school seats in Flanders (Belgium)

Supervisor: Tom Demeulemeester
Email: tom.demeulemeester@maastrichtuniversity.nl
Discipline: Mathematical Economics, Operations Research

Description

In many cities around the world, a centralized algorithm decides how the scarce high-school seats are assigned to prospective students. In this thesis, you will investigate the performance of the mechanism that is currently used in Flanders (Belgium) on real-world data from the cities of Ghent and Antwerp in 2023 and 2025.

The algorithm that is used in Flanders is the Efficiency Adjusted Deferred Acceptance (EADA) mechanism, proposed by Kesten (2010). In essence, this mechanism will find Pareto-improvements upon the Deferred Acceptance mechanism that only violate priorities of students that are not affected by these improvements. Additionally, one can investigate how the EADA mechanism performs compared to other mechanism, such as the Deferred Acceptance mechanism itself, or other mechanisms that find Pareto-improvements upon the Deferred Acceptance algorithm.

References

Kesten, O. (2010). School choice with consent. The Quarterly Journal of Economics, 125(3), 1297-1348.

4.3 Empirical evaluation of the Price of Diversity in the Traveling Salesperson Problem

Supervisors: Tom Demeulemeester & Rui Jorge Almeida
Email: tom.demeulemeester@maastrichtuniversity.nl & RJ.Almeida@maastrichtuniversity.nl
Discipline: Operations Research

Description

In this thesis, you will empirically investigate the recent *Price of Diversity* concept that was proposed by de Berg et al. (2025).

In many settings, it may be interesting to return a set of diverse solutions, rather than a single solution. However, it is often not possible to return multiple solutions that are all optimal. Consider, for example, the Traveling Salesperson Problem (TSP). As a measure of diversity, one could require to return a set of solutions such that all solutions in this set are edge-disjoint (i.e., have no edges in common). The Price of Diversity then reflects the ratio between the objective value of the worst solution in that subset over the objective value of an optimal solution.

When selecting two edge-disjoint TSP tours in an instance where all vertices lie on a single line, it is known that the Price of Diversity is $\frac{13}{7}$. This means that, in any instance of this problem, it is always possible to select two edge-disjoint TSP tours such that the objective value of the worst tour is at most $\frac{13}{7}$ times the optimal objective value. In this thesis, you will evaluate empirically how this ratio behaves for different instances of the TSP when all cities lie on a single line, and report on whether it is common for this ratio to be near $\frac{13}{7}$, or whether it is often possible to select two edge-disjoint TSP tours without losing too much in terms of cost, with respect to the optimal.

References

de Berg, M., Martínez, A. L., & Spieksma, F. (2025). Disjoint Tours and the Price of Diversity. arXiv preprint arXiv:2507.13026.

4.4 Fairness in Scheduling

Supervisor: Valentin Michaux & Tjark Vredeveld
Email: valentin.michaux@maastrichtuniversity.nl & t.vredeveld@maastrichtuniversity.nl
Discipline: Operations Research

Description

In this thesis you will investigate the trade-off between efficiency and fairness in two/multi-agent scheduling problems through the concept of the Price of Fairness (PoF), the efficiency loss incurred when fair allocations are required. For example, a schedule that minimizes the sum of completion times may heavily disadvantage one agent by putting all of his jobs at the end of the schedule. On the other hand, a fair solution aims to balance agents' utilities, maximize the minimum utility over all agents, or even be proportionally fair over changes in utilities for all agents, potentially increasing the overall cost of the solution.
The project can be pursued in a theoretical, practical, or mixed direction. On the theoretical side, you may analyze structured variants of the problem, improve known bounds on the PoF for these subproblems, or design for them near-fair solution concepts with provably small efficiency loss. On the practical side, you will experimentally evaluate how fair common scheduling algorithms behave on realistic instances and develop improved algorithms or heuristics that reduce PoF for specific problem classes. The goal is to understand when fairness truly conflicts with efficiency and when it can be achieved almost for free in real scheduling settings.

References

Price of fairness in two-agent single-machine scheduling problems Agnetis A., Chen B., Nicosia G., & Pacifi A., 2019, European Journal of Operational Research.

Analyzing the price of fairness in scheduling problems with two agents Yu J., Liu P., Lu X., & Gu M., 2025, European Journal of Operational Research.

4.5 Patient Scheduling with Optimized Overbooking

Supervisor: Lissa Melis
Email: lissa.melis@maastrichtuniversity.nl
Discipline: Operations Research
Required Courses: EBC2121 Advanced Algorithms

Description

Outpatient clinics frequently experience patient no-shows and uncertain consultation durations. These uncertainties lead to idle physician time, reduced utilization, and financial losses. A common managerial response is overbooking: scheduling more patients than available time slots to compensate for expected no-shows.

However, excessive overbooking may result in long waiting times, staff overtime, and reduced patient satisfaction. Determining an appropriate overbooking strategy therefore requires balancing efficiency and service quality under uncertainty.

This thesis investigates how to optimize overbooking decisions in outpatient scheduling using quantitative Operations Research techniques. Note that the goal of this thesis is not to build a ML prediction model to estimate no-show probabilities. The focus will lie on the scheduling approach.

Two students can work on this topic, each focusing on a different methodological angle. This angle will be discussed during the first meeting with the supervisor. Possible research questions include:

- What is the optimal overbooking level given stochastic consultation durations?
- What is the optimal overbooking level given stochastic no-show probabilities?
- Can overbooking be improved by accounting for heterogeneous no-show risks across patient types?
- How can we create a robust overbooking policy, taking into account the hospital might misestimate the no-show rate?

References

Cayirli, T., & Veral, E. (2003). Outpatient scheduling in health care: a review of literature. Production and operations management, 12(4), 519-549.

Zeng, B., Turkcan, A., Lin, J., & Lawley, M. (2010). Clinic scheduling models with overbooking for patients with heterogeneous no-show probabilities. Annals of Operations Research, 178(1), 121-144.

4.6 The location routing problem for optimizing parcel locker locations

Supervisor: Lissa Melis
Email: lissa.melis@maastrichtuniversity.nl
Discipline: Operations Research
Required Courses: EBC2121 Advanced Algorithms

Description

A parcel locker is a secure, self-service storage unit where customers can pick up or drop off packages at their convenience, reducing the need for home deliveries. This thesis explores the Location Routing Problem (LRP) in the context of parcel lockers, aiming to determine the optimal placement of lockers while minimizing delivery costs and maximizing accessibility for customers. The study will integrate vehicle routing constraints to account for last-mile logistics efficiency. Using mathematical optimization techniques such as mixed-integer programming or heuristic algorithms, the research will analyze trade-offs between operational costs and customer convenience.

References

A simulated annealing algorithm for the vehicle routing problem with parcel lockers. Vincent, F. Y., Susanto, H., Jodiawan, P., Ho, T. W., Lin, S. W., & Huang, Y. T., 2022, IEEE Access, 10, 20764-20782.

Location-routing problem: a classification of recent research. Mara, S. T. W., Kuo, R. J., & Asih, A. M. S., 2021, International Transactions in Operational Research, 28(6), 2941-2983.

4.7 Evolution of Maximum Weighted Cliques in Dynamic Graphs: An Algorithmic Approach

Supervisor: Frank Phillipson
Email: f.phillipson@maastrichtuniversity.nl
Discipline: Operations Research
Required Courses: EBC2121 Advanced Algorithms

Description

This thesis explores the problem of tracking the maximum weighted clique in a dynamically evolving graph. Specifically, we study how the maximum weighted clique of an initial graph evolves when edge weights change incrementally over time. The research is motivated by applications in network analysis, optimization, and machine learning, where graphs often evolve dynamically. Using concepts from dynamic graph algorithms, we aim to develop efficient methods for updating the maximum weighted clique instead of recomputing it from scratch at each step.

Research Questions:

- How can dynamic graph algorithms be adapted to efficiently track the evolution of the maximum weighted clique in a changing graph?
- What are the computational trade-offs between recomputing the clique from scratch and incremental update methods?
- How does the structure of the graph (e.g., sparsity, density) affect the performance of different update strategies?

References

1 *Incremental K-clique clustering in dynamic social networks.* Duan, Dongsheng, et al., Artificial Intelligence Review 38 (2012): 129-147.

2 *Finding all maximal cliques in dynamic graphs.* Stix, Volker, Computational Optimization and applications 27 (2004): 173-186.

4.8 Improving Binary Optimization: A Duality-Driven Approach to Constraint Selection

Supervisor: Frank Phillipson
Email: f.phillipson@maastrichtuniversity.nl
Discipline: Operations Research
Required Courses: EBC2121 Advanced Algorithms

Description

Combinatorial optimization problems arise in various fields, from logistics and finance to machine learning and operations research. Many of these problems can be formulated as binary quadratic models, which are notoriously difficult to solve efficiently at scale. A common approach is to express these problems as Quadratic Unconstrained Binary Optimization (QUBO) formulations, making them suitable for heuristic and exact solvers. However, the efficiency of solving QUBOs heavily depends on how the problem is constructed. Large, dense formulations can slow down optimization algorithms, making it crucial to refine the problem structure while preserving solution quality.

This project explores ways to create QUBOs more efficiently by leveraging information from relaxed linear programs and classical dual formulations. Linear relaxations provide valuable insights into constraint importance, allowing us to selectively include or exclude constraints that contribute most to the optimal solution. By integrating duality principles, we can further refine how penalties are assigned in the QUBO formulation.

An additional layer of refinement comes from an iterative sampling approach. After generating an initial QUBO solution, we analyze patterns in the first set of results. If specific variables consistently take on binary values across multiple runs, we can fix them, reducing the problem size and re-optimizing. This iterative process gradually refines the problem while maintaining accuracy, ultimately leading to faster and more efficient optimization.

By combining these strategies, the project aims to enhance the efficiency of solving binary quadratic problems, making large-scale combinatorial optimization more tractable. The proposed methodology will be tested using both classical heuristic solvers and modern optimization frameworks to evaluate improvements in computational speed, solution quality, and problem sparsity.

References

1 *The quadratic unconstrained binary optimization problem.* Punnen, Abraham P., Springer International Publishing 10 (2022)

4.9 Statistical stopping rule for combinatorial optimisation

Supervisor: Frank Phillipson
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Discipline: Operations Research
Required Courses: EBC2121 Advanced Algorithms

Description

Quantum optimization methods such as quantum annealing and variational quantum algorithms (e.g., QAOA) are increasingly used to solve combinatorial optimization problems. In these settings, candidate solutions are obtained by repeated measurements ("shots") from a probabilistic quantum process. Each shot incurs non-negligible hardware cost and limited access time. Consequently, the main bottleneck is often the number of samples (shots) rather than classical computation time.

Standard classical metaheuristics and hybrid quantum–classical workflows typically rely on many repeated evaluations to identify high-quality solutions. However, many of these samples provide diminishing returns. This raises a fundamental question: ***Can we use statistical infer-***

ence to decide early when enough samples have been collected and further sampling is unlikely to significantly improve the solution?

This thesis project studies distribution-based statistical stopping rules that estimate the objective-value distribution of sampled solutions and use this information to terminate the search efficiently while maintaining solution quality.

Research Objectives: The student will:

- Study and compare several stopping strategies.
- Formulate statistical stopping criteria for sequential sampling.
- Integrate these criteria into simple search heuristics.
- Evaluate their effectiveness on QUBO optimization problems using simulated and quantum annealing.

References

1 *A tutorial on Bayesian optimization.* Frazier, Peter I., arXiv preprint arXiv:1807.02811 (2018).

2 *New two-stage and sequential procedures for selecting the best simulated system.* Chick, Stephen E.,Operations Research 49.5 (2001): 732-743.

4.10 Dynamic Kidney Exchange

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Required Courses: EBC2121 Advanced Algorithms

Description

Kidney exchange programmes have been established to swap incompatible patient-donor pairs of kidney-disease patients. In these programmes, if a donor for a patient is not compatible with the patient, one can try to swap with another pair such that they are cross-wise compatible, that is, patient of pair 1 is compatible with the donor of pair 2 and the patient of pair 2 is compatible with the donor of pair 1. Of course, these swaps can also be done in a multi-way setting with more patient-donor pairs involved. In these programmes there are timepoints at which an allocation of patients to donors are made such that every patient paired together with a donor that donates a kidney will receive one. If at such an allocation timepoint a patient does not receive a donor, their pair will automatically be transferred to the next allocation time, unless the pair withdraws.

In the simplest form, compatibility is based on the bloodtype, and their are 16 different types of patient-donor pairs (X, Y) with $X, Y \in \{A, B, O, AB\}$. Within two consecutive allocation times, the number of patient-donor pairs of type (X, Y) follows a Poisson-distribution λ_{XY}. Further, a patient-donor pair that is not matched in the current round, leaves with probability μ (independent of the type) and will not enter the next round. The goal is to find a matching procedure that minimizes the expected number of patient-donor pairs that leave without being matched.

The goal of this thesis is to find good matching procedures for the above described problem. One could start by considering a simplified case in which the bloodtype is either A or B, or the case when these are only A and O.

References

Dynamic Kidney Exchange, Ünver, 2010, Review of Economic Studies 77, pp. 372-414.

4.11 The vertex cover problem for regular graphs

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Required Courses: EBC2121 Advanced Algorithms, EBS2073 Introduction to Software in Operations Research

Description

The vertex cover problems is one of the most well-known NP-hard optimization problems. The unweighted version admits a very simple combinatorial 2-approximation. Other algorithms exist that are 2-approximations for the weighted version. It is widely believed that no better approximations exist. In a technical report Uriel Feige has shown that vertex cover is hardest to approximate on regular graphs. Carefully study the article by Feige to understand what is meant by the previous sentence. Demonstrate the construction he used with several examples. Then study the structure of several classes of regular graphs and their minimum vertex covers. Starting from this, the thesis can go into different directions: algorithms for special classes of regular graphs, analyzing graph classes for which finding vertex covers is especially difficult, etc. This is a theoretical topic. You have to like studying graphs and their structure if you want to work on this topic.

References

Vertex cover is hardest to approximate on regular graphs Uriel Feige, Technical Report, 2003. `http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.98.7630&rep=rep1&type=pdf`

4.12 The disease of graph theory

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Description

A famous open problem in graph theory is the graceful tree conjecture. Find out what that conjecture is about and review the known results and the history of that conjecture. Develop an

integer linear programming formulation for that problem and use it to test certain classes of trees for being truthful.

References

Chapters 1 and 2 of: *A Dynamic Survey of Graph Labeling* Joseph A. Gallian. `http://www.combinatorics.org/files/Surveys/ds6/ds6v20-2017.pdf`

4.13 The Traveling Tournament Problem

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Required Courses: EBC2121 Advanced Algorithms, EBS2073 Introduction to Software in Operations Research

Description

The Traveling Tournament Problem (TTP) is a sports timetabling problem, that seeks for a schedule of a round robin sports league satisfying several constraints. This is a very hard problem even for small instances. The goal of this thesis is to develop very quick heuristics for this problem, which can find schedules that may not be optimal or may not satisfy all the constraints. The first step is to define an appropriate objective function for evaluation. Another task is to develop an integer programming formulation that can solve at least very small instances. Further details can be specified depending on the chosen approaches, objectives, etc.

References

Easton, K., Nemhauser, G. & Trick, M. *The Traveling Tournament Problem. Description and Benchmarks.* In Principles and Practice of Constraint Programming — CP 2001, pages 580-584, Springer, 2001.

Minimum travel objective repository `https://robinxval.ugent.be/RobinX/travelRepo.php`

4.14 Impact of gender-neutral toilets on waiting times

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Required Courses: EBS2073 Introduction to Software in Operations Research

Description

The principal motivation for introducing gender-neutral toilets is to accomodate people who do not fit into the gender binary. Some claim gender-neutral toilets also improve average waiting times and/or fairness of waiting times between different groups. Of course, any such benefits would depend on a good implementation; Greed (2019) discusses a number of practical aspects.

In OR, the discipline of queueing theory (which you saw in the second-year course "Operations research") offers tools with which we could investigate such claims about waiting times. As one example: Huh, Lee, Park and Park (2018) did a simulation study of a toilet system based on a queueing-theoretical model.

In the thesis, you will compare two queueing systems with each other to see what is the impact of adding gender-neutral toilets. There are more or less two possible routes: either keep the systems very simple and analyse them theoretically, or study more realistic (so more complicated) systems with simulation methods.

References

Clara Greed (2019). Join the queue: including women's toilet needs in public space. *The Sociological Review Monographs*, volume 67 issue 4, pp. 908-926. `https://journals.sagepub.com/doi/10.1177/003`

Woonghee Tim Huh, Jaywon Lee, Heesang Park & Kun Soo Park (2018). The potty parity problem: towards gender equality at restrooms in business facilities. *Socio-Economic Planning Sciences*, volume 68, article 100666. `https://www.sciencedirect.com/science/article/pii/S00380121183`

4.15 Online max-flowtime scheduling

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Discipline: Operations Research
Required Courses: EBC2121 Advanced Algorithms, EBS2073 Introduction to Software in Operations Research

Description

In my master thesis (van Mill 2024) I analysed some algorithms for a particular scheduling problem. In this problem, the jobs arrive over time and are not known in advance; they have to be scheduled "online", meaning "on the go". There are multiple machines, each of which can have a different speed. The objective is to minimise the maximum "flowtime" among the jobs, where "flowtime" means waiting time plus processing time, i.e. the total time the job spends in the system.

My work left a number of open questions; among other things I proposed an algorithm called Semi-greedy, which seemed promising but for which I did not prove any performance guarantee. In your thesis, you will study some kind of algorithm for the problem; it can also be an algorithm that you come up with yourself. You can do simulation to investigate the performance of the algorithm, and maybe even prove a guarantee.

References

PJ van Mill (2024). *Simple online algorithms for maximum-flowtime scheduling on related machines* (MSc thesis, Maastricht University). `https://pjvm.nl.eu.org/academic/msc-thesis/`

4.16 Little clairvoyance in Scheduling

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Discipline: Operations Research
Required Courses: EBC2121 Advanced Algorithms

Description

Consider the classical problem of minimizing the total flow time of jobs on a single machine in the online setting where jobs arrive over time. It has long been known that the Shortest Remaining Processing Time (SRPT) algorithm is optimal when the job sizes are known up-front (Schrage, 1968). But in the non-clairvoyant setting where job sizes are revealed only when the job finishes, no algorithm can be constant-competitive (Motwani, Phillips, and Torng, 1994), meaning that we can find examples in which the algorithm performs extremely poorly compared to an adversary who knows all the processing times in advance.

Gupta et al. (2025) introduce the concept of ϵ-clairvoyance, which means that the remaining completion time of a job becomes known as soon as only an ϵ-fraction of the job is left to be processed. They show that there exists an algorithm, called SLF, that finds a solution that is at most an factor $\lceil 1/\epsilon \rceil$ away from the optimal solution value obtained by SRPT in the clairvoyant setting.

The goal of this thesis is to see how these algorithms work when the value of ϵ is randomly chosen from a specific distribution, or you can find your own algorithms to deal with this type of ϵ-clairvoyance.

References

A Little Clairvoyance Is All You Need, A. Gupta, H. Kaplan, A. Lindermayr, and S. Yingchareonthawornchai, 2025, Symposium on Foundations on Computer Science (FOCS).

Non-Clairvoyant Scheduling, R. Motwani, S.J. Phillips, and E. Torng, (1994), Theoretical Computer Science, 130 (1), pp. 17–47.

A Proof of the Optimality of the Shortest Remaining Processing Time Discipline, L. Schrage (1968), Operations Research 16 (3) pp. 687–690.

4.17 Lock scheduling

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Discipline: Operations Research
EBC2121 Advanced Algorithms

Description

Consider the fuel consumption of vessels when passing locks for inland waterway transportation. The goal of a single vessel is to minimize its fuel consumption while arriving on its destination in time. Given the current position of vessel j, its preferred time of arrival at the destination, a time d_j for the arrival at the lock can be computed which minimizes the fuel consumption for this vessel. As the fuel use is cubic in the velocity of the vessel, a deviation of the of the arrival time at the lock of δ time units, implies an increase in the fuel consumption in the order of δ^2. As teh fuel use differs between different vessels, the extra fuel needed is $w_j\delta^2$ for vessel j.

Assuming a lock capacity of 1 vessel, and an operation time of 1 time unit, the problem of minimizing the total fuel consumption over n vessels can be formulated by finding arrival times C_j for each vessel $j = 1, \ldots, n$ such that the differences between two arrival times needs to be at least 1. This problem can be seen as a scheduling problem of unit size jobs so as to minimize the total weighted squared deviation of job specific due dates.

If the job sizes are not restricted to be unit sized, this problem is known to be NP-hard. On the other hand, if all weights are the same, the problem can be solved in polynomial time.

The goal of this project is to come up with some nice algorithms that find good approximations for the (unit-size) scheduling problem to minimize the total weighted squared deviation from job specific due dates.

References

Vessel velocity decisions in inland waterway transportation under uncertainty, M.Y. Buchem, J.A.P. Golak, and A. Grigoriev, 2022, European journal of operational research 296 (2), 669-678.

Scheduling under weighted squared deviation from due dates for unit sized jobs, C. Koopmans, 2021, MSc thesis Leiden University, `https://studenttheses.universiteitleiden.nl/handle/1887/4171487`

Completion time variance minimization on a single machine is difficult., W. Kubiak, 1993, Operations Research Letters 14, 49-59.

4.18 Scheduling with rejections

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Required Courses: EBC2121 Advanced Algorithms (is a pre, not a must)

Description

One of the classical results in the field of scheduling is Smith's rule, which tells you how to minimize the sum of weighted completion times on a single machine. In the usual setting, you are required to schedule all n jobs on the machine. Suppose now that you need to schedule only a fixed number $k \leq n$ of jobs on the machine. The problem is to determine which jobs to schedule in order to minimize the sum of weighted completion times. Can you come up with an algorithm, some heuristics, a hardness result and/or present some experimental results by implementing a computer program?

In the problem of minimizing the sum of weighted completion times on a single machine, we are given n jobs which need to be scheduled on one machine. Each job has a given processing time p_j and a weight w_j and the machine can only process one job at a time. All jobs are available from the beginning. Smith (1956) showed that scheduling the jobs in non-decreasing order of their ratio of processing time over weight (w_j/p_j) is the optimal algorithm for this setting.

However, there are many applications in which you need not schedule *all* jobs on the machine, but you can decide not to schedule some of the jobs. In one variant, there is e.g. a rejection cost

for every job you decide not to schedule, and the objective is to minimize the sum of the weighted completion times of the scheduled jobs plus the rejection costs of the unscheduled jobs, see Engels et al. (2003).

In this thesis you will consider another variant, where jobs have no rejection costs, but you are given a fixed number $k \leq n$. The objective is to select a subset of exactly k jobs and minimize the sum of weighted completion times of these k jobs (the other jobs are simply disregarded). Smith's rule tells you the order in which you need to process these jobs, but it is unclear how to choose the subset of jobs that minimizes the objective function. In the case of equal weights, this problem can still be solved efficiently, namely, by selecting the k smallest jobs. However, when the weights can be arbitrary, it is still unknown what is the best selection.

In this thesis you will first do a literature review. Then you can start to think about developing an (approximation) algorithm or some heuristics. You are expected to implement these in a computer program and to test their quality.

References

Smith, W. E. (1956). Various optimizers for single-stage production. *Naval Research Logistics Quarterly*, 3(1–2), 59–66.

Engels, D. W., Karger, D. R., Kolliopoulos, S. G., Sengupta, S., Uma, R. N., & Wein, J. (2003). Techniques for scheduling with rejection. *Journal of Algorithms*, 49(1), 175–191.